PEDRO LIZÁRRAGA BRAMBILA

Senior Software Engineer | Full-Stack | TypeScript, Node.js, React, Angular & GraphQL
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Professional Summary

Pedro is a Senior Software Engineer with 15+ years of experience building scalable, reliable full-stack web applications across healthcare, e-commerce, finance, transportation, and government. He works with TypeScript, Node.js, NestJS, React, Angular, GraphQL, and PostgreSQL on AWS and Azure, integrates Epic FHIR APIs, and holds the Apollo Graph Developer – Associate certification. He also builds AI agents with the Vercel AI SDK, conducts code reviews, and mentors engineers in cross-functional Agile teams.

Technical Skills

- **Languages & Frameworks:** TypeScript, JavaScript, Node.js, NestJS, Express.js, React, Next.js, Angular, Hono, Python, Ruby on Rails, C#, ASP.NET/.NET
- **APIs & Architecture:** GraphQL, Apollo Server, Apollo Federation, REST, system design, software architecture, design patterns
- **Databases:** PostgreSQL, MySQL, SQL Server, Oracle, Cloudflare D1, Prisma, Drizzle
- **Cloud:** AWS Lambda, Amazon S3, Microsoft Azure (Azure AI Search, Azure Blob Storage), Cloudflare Workers
- **AI Engineering:** AI agents, LLM applications, OpenAI APIs, Vercel AI SDK, tool calling, RAG, workflow automation
- **Testing & Delivery:** Jest, Vitest, Playwright, Testcontainers, Docker, GitHub Actions, Bitrise, Azure DevOps, Git, CI/CD, Agile, Scrum, Kanban, Jira
- **Healthcare:** HL7 FHIR, Epic FHIR, healthcare data transformation, mapping, and normalization
- **Languages:** Spanish, English (B2+)

Professional Experience

Freelance Software Engineer | Independent

Remote, Mexicali, Baja California, Mexico | May 2022–Present

AI Appointment Agent for a Dental Clinic (in development) | Oct 2026–Present

- Developing an AI agent that answers patient questions and books, reschedules, and cancels appointments for a dental clinic over WhatsApp, with a team panel to take over conversations.
- Implemented tool calling and structured output with the Vercel AI SDK, code-enforced guards, and an evals suite.
- Built the backend on Cloudflare Workers, D1, and Hono and the panel with React, with Vitest tests and CI.
- Designed the solution as a reusable template, one deployment per client, to adapt it to other business niches.

Environment: TypeScript, Vercel AI SDK, OpenAI APIs, Cloudflare Workers, Durable Objects, D1, Drizzle, Hono, WhatsApp Cloud API, Better Auth, React, Vitest, GitHub Actions

Senior Software Engineer | Publicis Sapient

Remote, Mexicali, Baja California, Mexico | May 2022–Aug 2026

- Conducted code reviews and mentored and onboarded engineers joining the projects.

Healthcare Platform | Mar 2024–Aug 2026

- Built GraphQL and Node.js services in TypeScript for a patient-facing healthcare web and mobile app, owning the doctor search and details subgraph, system notifications, and caching in an Apollo Federation graph.
- Integrated Epic FHIR APIs (Practitioner, Patient, Appointment, DiagnosticReport, Observation, plus lab tests and medications), parsing source payloads into normalized domain models and joining Epic data with other subgraphs.
- Designed PostgreSQL models (notifications, preferences, Epic practitioner cache), GraphQL schemas, and resolver patterns.
- Strengthened delivery quality through Jest and Vitest test suites and cross-functional collaboration with frontend, mobile, QA, product, and engineering teams.
- Helped the React Native mobile team migrate Android and iOS pipelines from GitHub Actions to Bitrise.

Environment: TypeScript, Node.js, Apollo Server, Apollo Federation, GraphQL, PostgreSQL, Epic FHIR, Jest, Vitest, GitHub Actions, Bitrise

Document Search API | Sep 2023–Feb 2024

- Built a REST API with NestJS that enables search over proprietary documents using Azure AI Search semantic search and Azure Blob Storage.

AngularJS Migration | Sep 2022–Aug 2023

- Migrated an AngularJS (Angular 1) application to Angular 2+.

HR Reporting Tool | May 2022–Aug 2022

- Built a human-resources reporting tool with React and Vite.

Senior Frontend Developer | Fender Musical Instruments Corporation

Remote, Ensenada, Baja California, Mexico | Oct 2016–May 2022

- Built core experiences for Fender's dealer-facing B2B commerce platform, including product discovery, catalog, cart, ordering, and order-tracking workflows.
- Designed scalable Angular application structures and improved responsive behavior and navigation across key journeys.
- Connected frontend capabilities through GraphQL integrations and serverless Node.js services running on AWS Lambda.
- Collaborated with product, backend, and QA functions in Agile teams to plan, implement, test, and release customer-facing capabilities.
- Conducted code reviews and mentored and onboarded engineers on the team.

Environment: Angular, TypeScript, GraphQL, Node.js, Express.js, AWS Lambda, Amazon S3, Agile

Senior Full-Stack Developer | docDigitales

Mexicali, Baja California, Mexico | Aug 2014–Oct 2016

- Built an electronic-invoicing web platform covering invoicing, cost control, payroll, customer, and supplier workflows.
- Implemented full-stack functionality with Ruby on Rails, Python, JavaScript, Bootstrap, and MySQL, balancing usability with dependable business processing.

Full-Stack Developer | Secretaría de Seguridad Pública

Mexicali, Baja California, Mexico | Apr 2012–Jul 2014

- Led development of SAFyS, a web application used to administer and track public-security funds and subsidies.
- Automated monthly accounting and reporting activities using C#, ASP.NET/.NET, JavaScript, SQL Server, and Oracle.

Software Developer | Grupo GM Transport

Mexicali, Baja California, Mexico | Sep 2011–Mar 2012

- Developed C# and ASP.NET administrative systems supporting transportation operations and back-office processes.

Software Developer | Seres Soluciones en Software

Mexicali, Baja California, Mexico | Mar 2010–Aug 2011

- Developed client-server business applications with WinDev, WebDev, MySQL, and HyperFile Classic.

Selected Projects

starter-template-nestjs | github.com/plizarraga/starter-template-nestjs | 2026

- Created a production-ready NestJS 12 starter with Better Auth, PostgreSQL-backed sessions, RBAC, user administration, Prisma, Pino, and Vitest with Testcontainers, to bootstrap new backend projects with consistent best practices.

finance-tracker | github.com/plizarraga/finance-tracker | Jan 2026

- Built a mobile-first personal finance tracker with Next.js (App Router), Prisma, PostgreSQL, Better Auth, and automated tests with Vitest and Playwright.

RAG Chat over PDFs (personal MVP) | Jul 2024

- Built an MVP that answers questions over uploaded PDFs using NestJS, PostgreSQL, OpenAI embeddings and chat completions, and ChromaDB.

Certifications

- Modern GraphQL with Node – Complete Developers Guide | Udemy | Jul 2024
- Apollo Graph Developer – Associate | Apollo GraphQL | Jun 2024
- TypeScript: Your Complete Guide and Handbook | Udemy
- Design Patterns in JavaScript and TypeScript | Udemy
- Clean Code and SOLID Principles | Udemy

Education

Bachelor of Engineering, Computer Systems Engineering | Instituto Tecnológico de Mexicali | Mexicali, Mexico | 2005–2009